



DATA CLEANING IN EXCEL

Yuly Artami Sagala

➤ Start Slide

What is Data Cleaning?

Data cleaning is the process of identifying and correcting (or removing) errors and inconsistencies in data to improve its quality

Why is it important?

Clean data is essential for accurate analysis, better decision-making, and reliable business insights

Project Overview



Dataset Source

Sample Data Cleaning project from Alex The Analyst's YouTube Channel

Objective

To clean and organize raw sales data for accurate analysis and better decision-making.

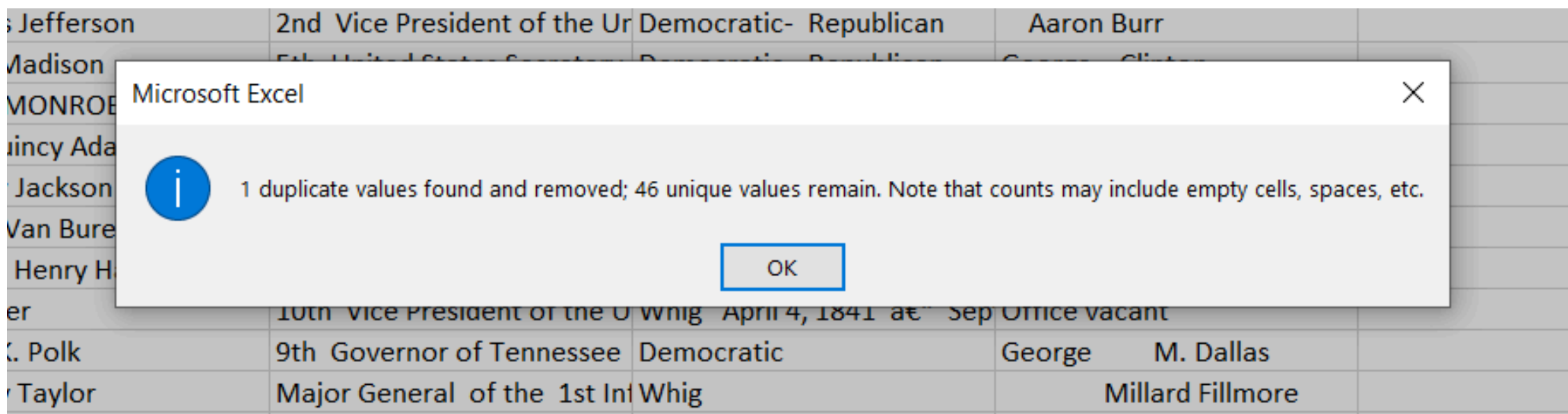
Dataset

	A	B	C	D	E	F	G	H	I	
1		S.No.	president	prior	party	vice	salary	date updated	date created	
2	0		1 George Washington	Commander-in-Chief of the	Nonpartisan	John Adams	\$5.000,00	14/07/2021	04/03/2012	
3	1		2 john adams	1st Vice President of the Uni	Federalist	Thomas Jefferson	\$10.000,00	14/07/2021	04/03/2012	
4	2		3 Thomas Jefferson	2nd Vice President of the Ur	Democratic- Republican	Aaron Burr	\$15.000,00	14/07/2021	04/03/2012	
5	3		4 James Madison	5th United States Secretary	Democratic- Republican	George Clinton	\$20.000,00	14/07/2021	04/03/2012	
6	4		5 JAMES MONROE	7th United States Secretary	Democratic- Republican	Daniel D. Tompkins	\$25.000,00	14/07/2021	04/03/2012	
7	5		6 John Quincy Adams	8th United States Secretary	Democratic- Republican	John C. Calhoun	\$30.000,00	14/07/2021	04/03/2012	
8	6		7 Andrew Jackson	U.S. Senator (Class 2) fro	Democratic	John C. Calhoun	\$35.000,00	14/07/2021	04/03/2012	
9	7		8 Martin Van Buren	8th Vice President of the Un	Democratic	Richard Mentor Johnson	\$40.000,00	14/07/2021	04/03/2012	
10	8		9 William Henry Harrison	United States Minister to Col	Whig	John Tyler	\$45.000,00	14/07/2021	04/03/2012	
11	9		10 john tyler	10th Vice President of the U	Whig April 4, 1841 â€“ Sep	Office vacant	\$50.000,00	14/07/2021	04/03/2012	
12	10		11 James K. Polk	9th Governor of Tennessee	Democratic	George M. Dallas	\$55.000,00	14/07/2021	04/03/2012	
13	11		12 Zachary Taylor	Major General of the 1st Inf	Whig	Millard Fillmore	\$60.000,00	14/07/2021	04/03/2012	
14	12		13 Millard Fillmore	12th Vice President of the U	Whig	Office vacant	\$65.000,00	14/07/2021	04/03/2012	
15	13		14 Franklin Pierce	Brigadier General of the 9th	Democratic	William R. King	\$75.000,00	14/07/2021	04/03/2012	
16	14		15 James Buchanan	United States Minister to the	Democratic	John C. Breckinridge	\$85.000,00	14/07/2021	04/03/2012	
17	15		16 Abraham Lincoln	U.S. Representative for Illin	Republican	Hannibal Hamlin	\$95.000,00	14/07/2021	04/03/2012	
18	16		17 Andrew Johnson	16th Vice President of the U	Democratic	Office vacant	\$105.000,00	14/07/2021	04/03/2012	
19	17		18 Ulysses S. Grant	Commanding General of the	Republican	Schuyler Colfax	\$115.000,00	14/07/2021	04/03/2012	
20	18		19 Rutherford B. Hayes	29th & 32nd Governor of Oh	Republican	William A. Wheeler	\$125.000,00	14/07/2021	04/03/2012	

The dataset contains various quality issues such as duplicates, missing values, and inconsistent formatting.

Removing Duplicate Values

Duplicate records can lead to inaccurate summaries and analysis.



Steps:

- Go to the Data tab
- Click Remove Duplicate
- In the dialog box select the relevant columns

1 duplicate row was removed, the dataset now contains only unique entries.

Standardizing Text Format

In the President column, names had inconsistent capitalization (e.g., john adams, JAMES MONROE), which can affect grouping and readability.

	C	D	
	president		prio
1	George Washington	=PROPER(C2)	Corr
2	john adams	John Adams	1st
3	Thomas Jefferson	Thomas Jefferson	2nd
4	James Madison	James Madison	5th
5	JAMES MONROE	James Monroe	7th
6	John Quincy Adams	John Quincy Adams	8th
7	Andrew Jackson	Andrew Jackson	U.S.
8	Martin Van Buren	Martin Van Buren	8th
9	William Henry Harrison	William Henry Harrison	Unit
10	john tyler	John Tyler	10th
11	James K. Polk	James K. Polk	9th
12	Zachary Taylor	Zachary Taylor	Maji
13	Millard Fillmore	Millard Fillmore	12th
14	Franklin Pierce	Franklin Pierce	Brig
15	James Buchanan	James Buchanan	Unit
16	Abraham Lincoln	Abraham Lincoln	U.S.
17	Andrew Johnson	Andrew Johnson	16th
18	Ulysses S. Grant	Ulysses S. Grant	Corr
19	Rutherford B. Hayes	Rutherford B. Hayes	29th
20	James A. Garfield	James A. Garfield	U.S.
21	Chester A. Arthur	Chester A. Arthur	20th
22	Grover Cleveland	Grover Cleveland	28th
23	Benjamin Harrison	Benjamin Harrison	U.S.
24	Grover Cleveland	Grover Cleveland	22nd

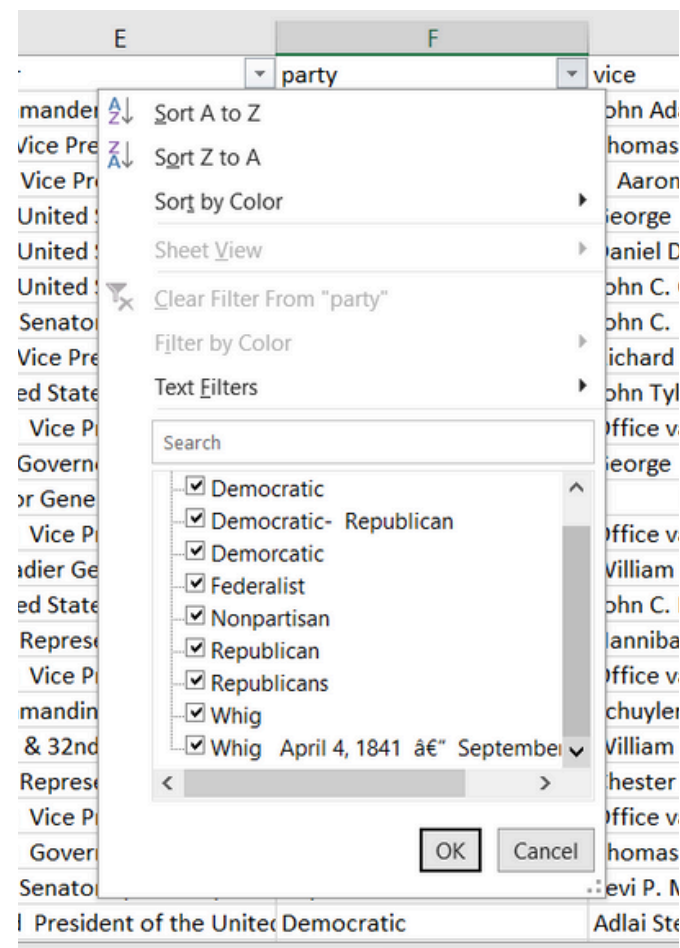
Steps:

- Insert a new column
- use the formula
=PROPER([cell])
- Drag the formula down to apply it to all rows
- Copy and paste as values to replace the original column

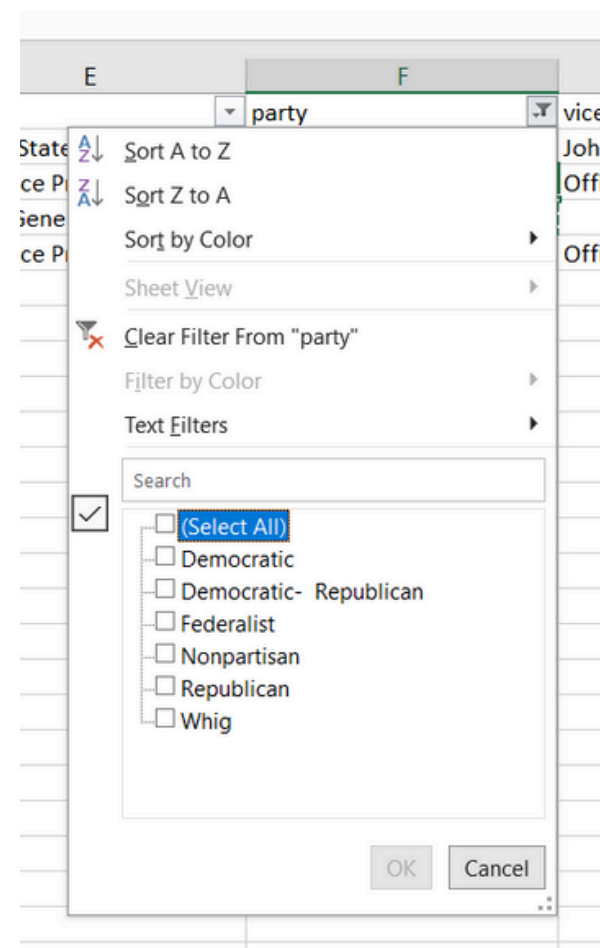
Fixing Inconsistent Category

The party column contained inconsistent or misspelled entries. these inconsistencies cause errors in grouping, filterting, and analysis.

Before



After



Steps:

- Filter the column to check for inconsistent values
- For this case, manually edit the entries to standardize
- Or we can use Find and Replace

Removing Extra Spaces

The Vice column contained leading, trailing, or multiple spaces between words, which can interfere with matching, filtering, or lookups.

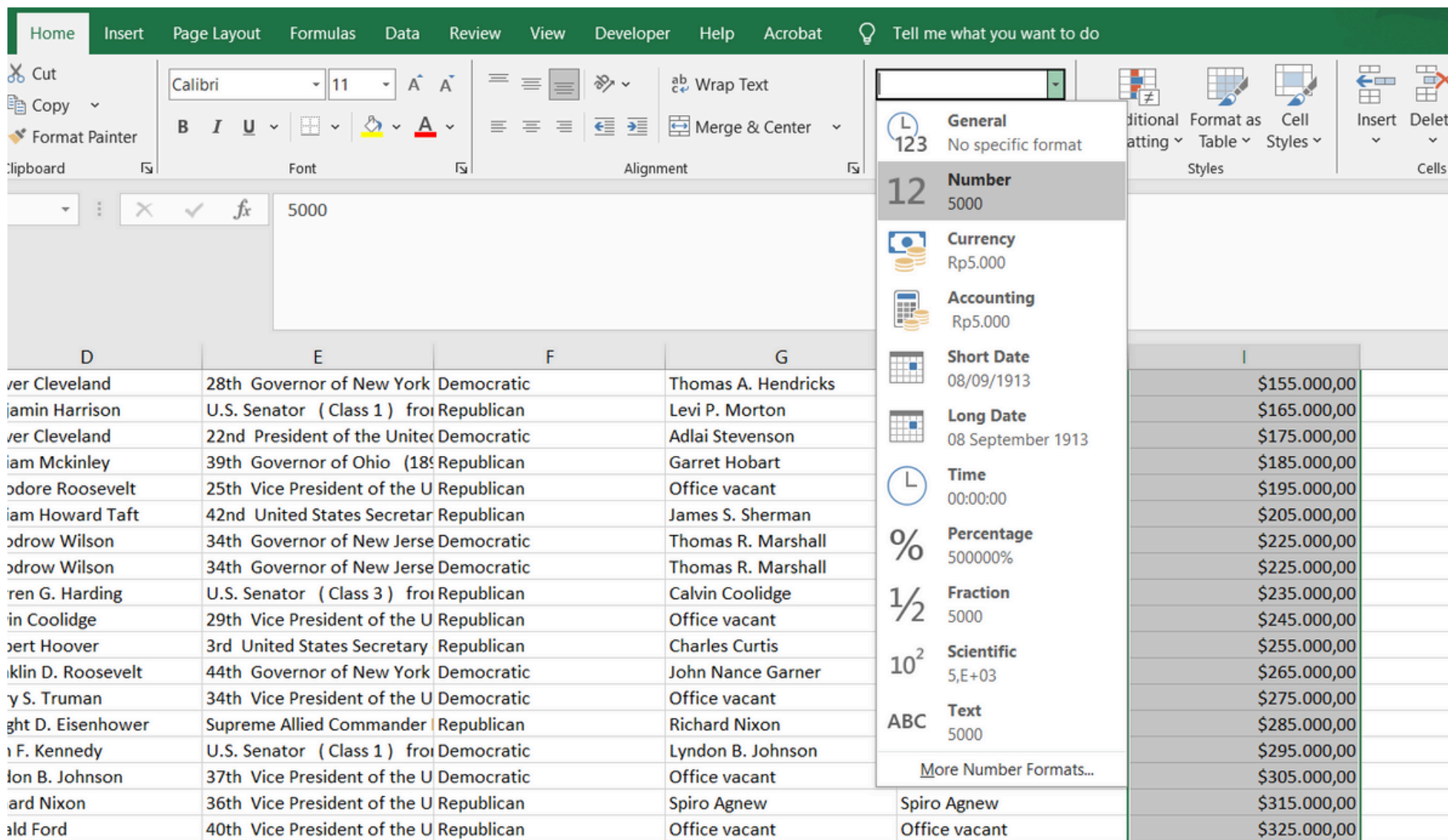
	G	H	
	vice		sala
	John Adams	=TRIM(G2)	
	Thomas Jefferson	Thomas Jefferson	
	Aaron Burr	Aaron Burr	
	George Clinton	George Clinton	
	Daniel D. Tompkins	Daniel D. Tompkins	
	John C. Calhoun	John C. Calhoun	
	John C. Calhoun	John C. Calhoun	
	Richard Mentor Johnson	Richard Mentor Johnson	
	John Tyler	John Tyler	
	Office vacant	Office vacant	
	George M. Dallas	George M. Dallas	
	Millard Fillmore	Millard Fillmore	
	Office vacant	Office vacant	
	William R. King	William R. King	
	John C. Breckinridge	John C. Breckinridge	
	Hannibal Hamlin	Hannibal Hamlin	
	Office vacant	Office vacant	
	Schuyler Colfax	Schuyler Colfax	
	William A. Wheeler	William A. Wheeler	
	Chester A. Arthur	Chester A. Arthur	
	Office vacant	Office vacant	
	Thomas A. Hendricks	Thomas A. Hendricks	

Steps:

- Insert a new column
- use the formula
`=TRIM([cell])`
- Drag the formula down to apply it to all rows
- Copy and paste as values to replace the original column

Changing Data Type

The Salary column was in currency format, which is treated as text in some tools like SQL, causing errors in numeric operations and filtering.



The screenshot shows the Microsoft Excel interface with the 'Home' tab selected. The 'Number' format dropdown menu is open, displaying various options. The 'Number' option is highlighted, showing the value '5000'. The background shows a table with columns D, E, F, and G. The table contains names and titles of U.S. Presidents and Vice Presidents, along with their political affiliations. The 'Salary' column (I) is visible on the right, showing values in currency format (e.g., \$155.000,00).

D	E	F	G
ver Cleveland	28th Governor of New York	Democratic	Thomas A. Hendricks
jamin Harrison	U.S. Senator (Class 1)	Republican	Levi P. Morton
ver Cleveland	22nd President of the United States	Democratic	Adlai Stevenson
iam McKinley	39th Governor of Ohio (1896)	Republican	Garret Hobart
odore Roosevelt	25th Vice President of the United States	Republican	Office vacant
iam Howard Taft	42nd United States Secretary of War	Republican	James S. Sherman
odrow Wilson	34th Governor of New Jersey	Democratic	Thomas R. Marshall
odrow Wilson	34th Governor of New Jersey	Democratic	Thomas R. Marshall
ren G. Harding	U.S. Senator (Class 3)	Republican	Calvin Coolidge
in Coolidge	29th Vice President of the United States	Republican	Office vacant
bert Hoover	3rd United States Secretary of War	Republican	Charles Curtis
iklin D. Roosevelt	44th Governor of New York	Democratic	John Nance Garner
y S. Truman	34th Vice President of the United States	Democratic	Office vacant
ght D. Eisenhower	Supreme Allied Commander in Europe	Republican	Richard Nixon
l F. Kennedy	U.S. Senator (Class 1)	Democratic	Lyndon B. Johnson
lon B. Johnson	37th Vice President of the United States	Democratic	Office vacant
ard Nixon	36th Vice President of the United States	Republican	Spiro Agnew
ald Ford	40th Vice President of the United States	Republican	Office vacant

Steps:

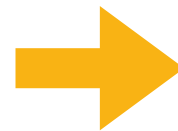
- Select the salary column
- Go to the Home tab - Number group
- Change the format to Number

Standardizing Date Format

The Date column contained inconsistent formats (e.g., 14/07/2021, 14 July 2021), which can cause sorting errors and confusion during analysis.

Before

	J	K
	date updated	date created
275000	14/07/2021	04/03/2012
285000	14/07/2021	04/03/2012
295000	14/07/2021	04/03/2012
305000	14/07/2021	04/03/2012
315000	14/07/2021	04/03/2012
325000	14/07/2021	04/03/2012
335000	14/07/2021	04/03/2012
345000	14/07/2021	04/03/2012
355000	14/07/2021	04/03/2012
365000	14/07/2021	04/03/2012
375000	14/07/2021	04/03/2012
395000	14 July 2021	01 February 2020
405000	14 July 2021	01 February 2020



After

	J	K
	date updated	date created
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	04/03/2012
10	14/07/2021	01/02/2020
10	14/07/2021	01/02/2020

Steps:

- Select the date column
- Go to the Home tab - Number group
- Click the dropdown and choose Short Date format

Redudant Data Removed

	A	B	C	D	E	F	G	H	
1	No	president	prior	party	vice	salary	date updated	date created	
2	1	George Washington	Commander-in-Chief of the	Nonpartisan	John Adams	5000	14/07/2021	04/03/2012	
3	2	John Adams	1st Vice President of the Uni	Federalist	Thomas Jefferson	10000	14/07/2021	04/03/2012	
4	3	Thomas Jefferson	2nd Vice President of the Un	Democratic- Republican	Aaron Burr	15000	14/07/2021	04/03/2012	
5	4	James Madison	5th United States Secretary	Democratic- Republican	George Clinton	20000	14/07/2021	04/03/2012	
6	5	James Monroe	7th United States Secretary	Democratic- Republican	Daniel D. Tompkins	25000	14/07/2021	04/03/2012	
7	6	John Quincy Adams	8th United States Secretary	Democratic- Republican	John C. Calhoun	30000	14/07/2021	04/03/2012	
8	7	Andrew Jackson	U.S. Senator (Class 2) fro	Democratic	John C. Calhoun	35000	14/07/2021	04/03/2012	
9	8	Martin Van Buren	8th Vice President of the Un	Democratic	Richard Mentor Johnson	40000	14/07/2021	04/03/2012	
10	9	William Henry Harrison	United States Minister to Col	Whig	John Tyler	45000	14/07/2021	04/03/2012	
11	10	John Tyler	10th Vice President of the U	Whig	Office vacant	50000	14/07/2021	04/03/2012	
12	11	James K. Polk	9th Governor of Tennessee	Democratic	George M. Dallas	55000	14/07/2021	04/03/2012	
13	12	Zachary Taylor	Major General of the 1st Inf	Whig	Millard Fillmore	60000	14/07/2021	04/03/2012	
14	13	Millard Fillmore	12th Vice President of the U	Whig	Office vacant	65000	14/07/2021	04/03/2012	
15	14	Franklin Pierce	Brigadier General of the 9th	Democratic	William R. King	75000	14/07/2021	04/03/2012	
16	15	James Buchanan	United States Minister to the	Democratic	John C. Breckinridge	85000	14/07/2021	04/03/2012	
17	16	Abraham Lincoln	U.S. Representative for Illin	Republican	Hannibal Hamlin	95000	14/07/2021	04/03/2012	

Steps:

- Removed column A
- Removed helper column for president and vice to avoid duplication.
- Removed the prior column that contained irrelevant information

Clean Data Ready!

	A	B	C	D	E	F	G	H	I
1	No	president	prior	party	vice	salary	date updated	date created	
2	1	George Washington	Commander-in-Chief of the	Nonpartisan	John Adams	5000	14/07/2021	04/03/2012	
3	2	John Adams	1st Vice President of the Uni	Federalist	Thomas Jefferson	10000	14/07/2021	04/03/2012	
4	3	Thomas Jefferson	2nd Vice President of the Ur	Democratic- Republican	Aaron Burr	15000	14/07/2021	04/03/2012	
5	4	James Madison	5th United States Secretary	Democratic- Republican	George Clinton	20000	14/07/2021	04/03/2012	
6	5	James Monroe	7th United States Secretary	Democratic- Republican	Daniel D. Tompkins	25000	14/07/2021	04/03/2012	
7	6	John Quincy Adams	8th United States Secretary	Democratic- Republican	John C. Calhoun	30000	14/07/2021	04/03/2012	
8	7	Andrew Jackson	U.S. Senator (Class 2) froi	Democratic	John C. Calhoun	35000	14/07/2021	04/03/2012	
9	8	Martin Van Buren	8th Vice President of the Un	Democratic	Richard Mentor Johnson	40000	14/07/2021	04/03/2012	
10	9	William Henry Harrison	United States Minister to Col	Whig	John Tyler	45000	14/07/2021	04/03/2012	
11	10	John Tyler	10th Vice President of the U	Whig	Office vacant	50000	14/07/2021	04/03/2012	
12	11	James K. Polk	9th Governor of Tennessee	Democratic	George M. Dallas	55000	14/07/2021	04/03/2012	
13	12	Zachary Taylor	Major General of the 1st Inf	Whig	Millard Fillmore	60000	14/07/2021	04/03/2012	
14	13	Millard Fillmore	12th Vice President of the U	Whig	Office vacant	65000	14/07/2021	04/03/2012	
15	14	Franklin Pierce	Brigadier General of the 9th	Democratic	William R. King	75000	14/07/2021	04/03/2012	
16	15	James Buchanan	United States Minister to th	Democratic	John C. Breckinridge	85000	14/07/2021	04/03/2012	
17	16	Abraham Lincoln	U.S. Representative for Illin	Republican	Hannibal Hamlin	95000	14/07/2021	04/03/2012	
18	17	Andrew Johnson	16th Vice President of the U	Democratic	Office vacant	105000	14/07/2021	04/03/2012	
19	18	Ulysses S. Grant	Commanding General of the	Republican	Schuyler Colfax	115000	14/07/2021	04/03/2012	
20	19	Rutherford B. Hayes	29th & 32nd Governor of Oh	Republican	William A. Wheeler	125000	14/07/2021	04/03/2012	
21	20	James A. Garfield	U.S. Representative for Ohi	Republican	Chester A. Arthur	135000	14/07/2021	04/03/2012	
22	21	Chester A. Arthur	20th Vice President of the U	Republican	Office vacant	145000	14/07/2021	04/03/2012	
23	22	Grover Cleveland	28th Governor of New York	Democratic	Thomas A. Hendricks	155000	14/07/2021	04/03/2012	
24	23	Benjamin Harrison	U.S. Senator (Class 1) froi	Republican	Levi P. Morton	165000	14/07/2021	04/03/2012	
25	24	Grover Cleveland	22nd President of the Unite	Democratic	Adlai Stevenson	175000	14/07/2021	04/03/2012	

Final Outcomes:

- ✓ Dataset cleaned and simplified
- ✓ No duplicates
- ✓ Consistent formatting (text, dates, numbers)
- ✓ Ready for further analysis in Excel, SQL, or Power BI

Now, this dataset is clean, structured, and analysis-ready.

THANK YOU



Yuly Artami Sagala



<https://github.com/yulyartamisagala>

➤ End Slide